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Cicada species recognition based on acoustic signals using dynamic time warping-graph based GraphMix, graph convolution network

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Abstract

Cicadas, known for their distinctive acoustic signals, have been subjects of classification research for years. Recent researches elaborated the species composition as effect of climate change, further raising the need of effective classification system. Traditional methods rely on manual classification by domain experts, while recent trends favor Artificial Intelligence (AI)-assisted approaches due to their efficiency. However, image-based recognition faces challenges due to cicadas' varied appearances and environmental factors. Deep learning approaches, particularly utilizing Mel-frequency cepstral coefficients (MFCC) spectrograms, have been effective but are limited by dataset size. Graph Neural Networks (GNN) have surfaced as a promising alternative, leveraging graph represen- tations to provide additional information like data relationships. In this study, we address the challenge of efficient classification with a small dataset while maximizing feature representation. We explore the effectiveness of MFCC and Chromagram features in a noisy environment, constructing unique graphs for each. Dynamic Time Warping (DTW) is employed to establish connec- tions between nodes. Our experiments on the cicada audio dataset demonstrate the superiority of Chromagram over MFCC, with graph-based approaches outperforming graph-less methods such as Recurrent Neural Networks (RNN). Our findings suggest the potential of graph neural networks in audio classification tasks and contribute to advancing the field's methodologies.

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Keywords: Insect Classification; Audio; Acoustic Signal; Speech Recognition; Graph Representation; GAT; GraphMix; GCN; RNN;

1. Introduction

Climate change is one of the most complex challenges of our time, affecting weather patterns and ecosystem worldwide. Monitoring these changes is critical, and cicada can serve as valuable indicators due to their behavior

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responsiveness to temperature shifts. Diversity loss and species composition changes have been shown to be the effects of such event [1]. Thus, researchers have been observing cicada population density of each cyclical species. Doing such a task manually is resource extensive and time consuming. Therefore, a cicada species recognition system is needed.

Cicadas are a type of insect well known for their distinctive loud acoustic signals, which serve as their communication media. Male cicadas can generate loud clicking noises reaching frequencies as high as 7 kHz with a sound pressure level of 100 db. These sounds are made from the vibration of membranes known as timbals, located on both side of their bodies. On the contrary, female cicadas generate less intense sound by using their wings to responds to the males' calling [2].

The research of cicada classification has been going on for a long time. The methods which are used can be grouped into two methods. The first one is manual classification by domain expert. The second one is Artificial Intelligence-assisted approach. The latter has seen more use recently due to its efficiency in both time and effort. Acoustic signal is preferred over images due to difficulties in capturing vivid details from complex environments. In addition, insects undergo certain phases that may alter their appearances, further reducing image-based recognition's effectiveness. [3]

The deep learning approach is usually done by transforming the waveform to its MFCC spectrogram. The spectrogram acts as an input for Convolutional Neural Network [4]. However, its performance tends to fall as the dataset size decreases. With a small pool of dataset containing only images and labels, the model will not offer promising results. To handle this, the data must contain enough information and excellent feature representation.

Lately, graph neural networks (GNN) have been gaining popularity and usage. GNN utilise graph representation as additional information that is processed with the input features. Graph representation is a method to transform dataset to nodes and edges. Depending on the task, the node can either act as a single representation of an instance or a segment in the instance. But what makes the graph representation powerful is the ability to capture both arbitrary and calculated relationships between the data. A well-structured graph addresses the issue related to data numbers.

Recent works such as graph representation learning for audio and music genre classification [5] and self-supervised with limited labeled audio data [6] have shown promising results. Each research focuses on capturing relationship in node level graph. However, research that uses graph approach to model audio data are limited. The possibilities that the current state of this field's results having reached full potential are close to abysmal.

In this work, we focus on the problem of insects in regards to efficiency in small number of data, while maximizing the representation capability of the features. As insect hunting and study by the researchers often take time in the forest, there may be a lot of noises made by surrounding environments. Wan et al performed research on the same dataset, found and removed a lot of noises above 10kHz [2] . Thus, we decided to test two features, MFCC and Chromagram. Chromagram has shown to be more noise-resilient over MFCC [7]. A unique graph is made for each of these features. To construct connection between nodes, we utilised Dynamic Time Warping algorithm. DTW is an algorithm for measuring similarity between two temporal sequences. Recent usage of DTW in KNN clustering for volcanic seismic waveform [8] have shown promising results. We will test graph-based approaches performance against graph-less approach that is popular in time series domain, RNN.

2. Previous Works

Tey et al. [2] focused on data representation through three variations of Ha¨rma¨ syllable segmentation. The data, which are spectrogram image from Fast Fourier Transform, are transformed to these segmentation variations. The spectrogram images are then used to train a simple convolutional neural network model. However, based on the provided accuracy graph, the improved Ha¨rma¨ did not perform any better than the original Ha¨rma¨ set.

Arpitha et al. [3] used the sound of wingbeats to classify four mosquito spefcies. The dataset was fetched from kaggle, containing 1200 recordings. On the pre-processing stage, Hamming window function was used to frame the signals. Then, octave scale is applied to scale the feature. On this stage, the spectrograms from mel scale and octave scale are compared. Spectrogram made with octave scale worked better. The researcher developed ResNet-50 which focuses on unsupervised transfer learning. This experiment proposed to develop a CNN model with unsupervised transfer learning strategies. With K-Fold validation an average accuracy of 96% was achieved. The researcher suggests that the model may be viable for other type of sound classification due to such result.

Zhang et al. [4] experimented with insect sound classification using MFCC as feature. The dataset was fetched from sound libraries known as ARS and USDA. Normalization and pre-emphasis is applied before extracting the mel scale. A CNN model was built to classify nine classes from 93 recordings. The method reached an average accuracy of 92.56% because the signal-to-noise ratio was balanced by pre-emphasis. This is expected according to [7], that MFCC needs special care in processing the noise to gain good result. Unfortunately, imbalanced number of the sound recordings resulted in some class achieving lower accuracy of 69%.

Dokania et al. [5] showed that it is possible to solve audio classification problem with graph-based approach. Siamese neural network is used to find similarities between music recordings. The research uses two datasets GTZAN and AudioSet. GTZAN contains 1000 audio samples, 30 seconds each while AudioSet contains 2084320 samples, 10 seconds each. The proposed graph approach reached 99.5% accuracy in GTZAN dataset and 91.3% in AudioSet.

Yu et al. [7] experimented with mfcc and chromagram as distance measurement features in several signal to noise ratio. On 15db distance measure using mfcc scored 83.4% accuracy, 15% lower than chromagram's performance. On 5db mfcc's performance suffered from a drastic loss of ±62% resulting in 21.5% accuracy. However chromagram managed to retain high accuracy regardless of the shift in SNR. Chromagram scored 98.4% , 97.8% on 5db, 15db SNR. Considering the environment where insect classification model may be used in real world, robustness to noise is a must-have.

Ida et al. [8] proposed the use of k-means clustering for classifying volcano-seismic events. The distance measure used is distance calculated by Dynamic Time Warping (DTW) algorithm. The DTW algorithm worked well with timeseries data. The use of DTW is preferable because it is directly applied to waveforms without any arbitrary assumptions. Thus, a less subjective result was given.

3. Methods

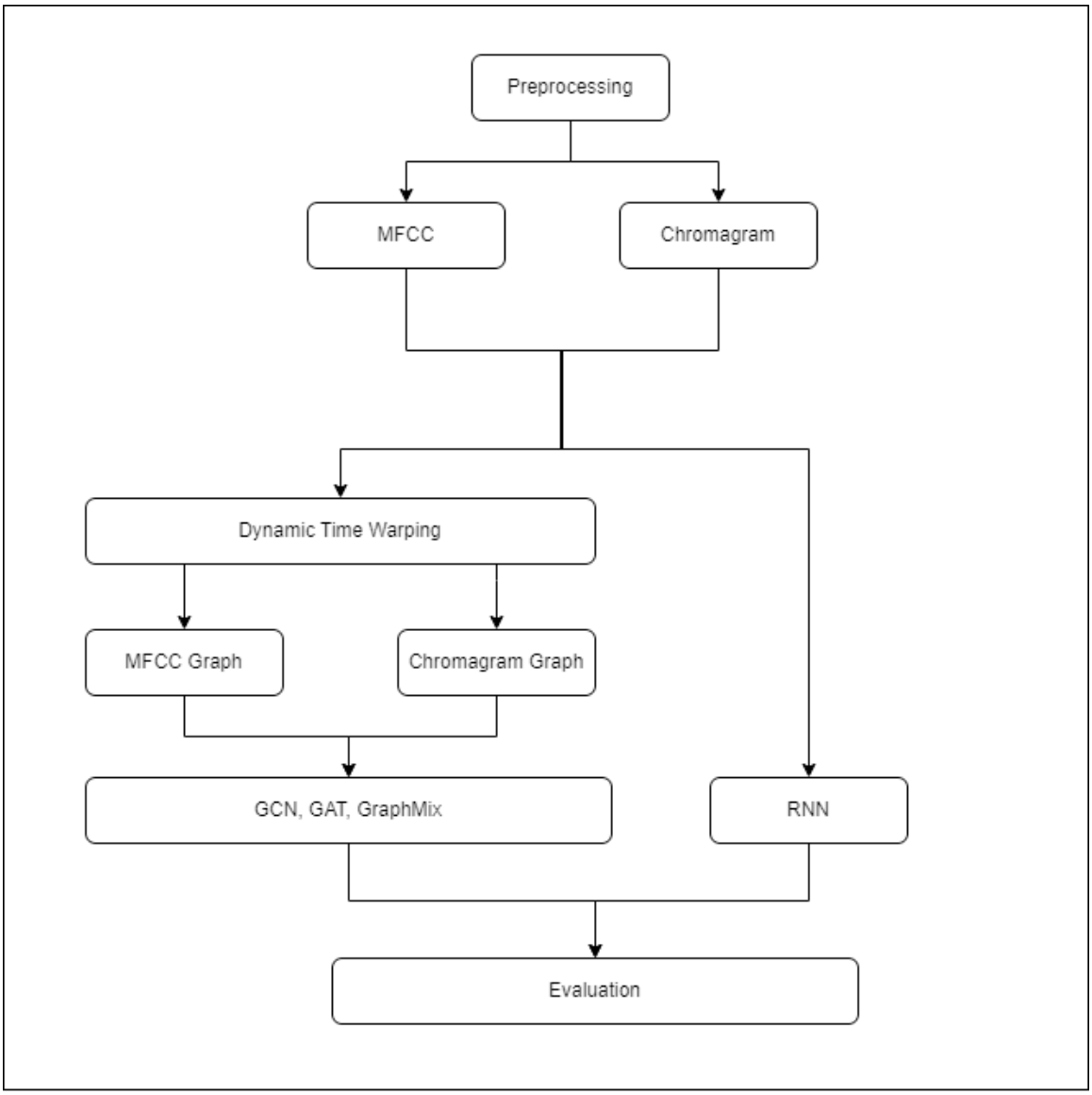


Fig. 1: The overview of our workflow

Fig 1 shows the overall process. First, data is pre-processed and extracted into two versions, namely mfcc and chromagram respectively. To evaluate how effective the graph approach is compared to other methods, we branched out a graph-less approach using Recurrent Neural Network. On the left side, both versions of dataset are transformed into graphs version by utilising Dynamic Time Warping. The graphs are then used to train the GNN models, each model providing results of mfcc graph and chromagram graph. On the right side, the raw mfcc and chromagram are used to train the RNN. Results from both approaches on both features will be shown below.

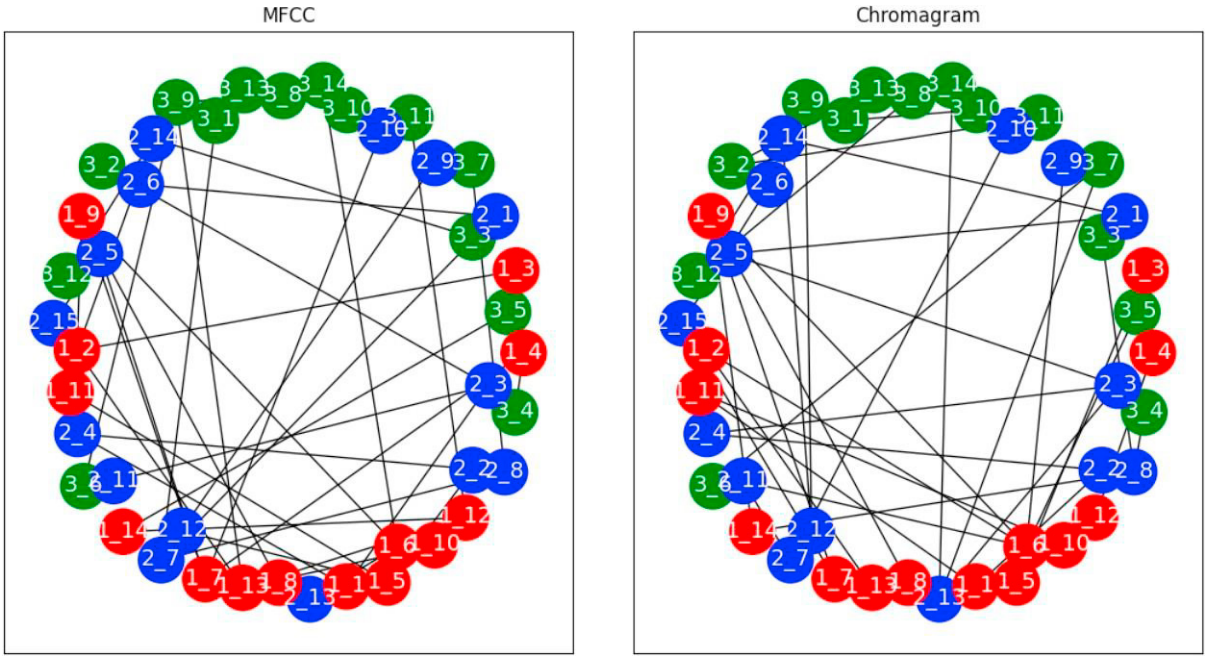
3.1. Data Preprocessing

We retrieved ”Cicada Species Detection Based on Acoustic Signals” dataset from kaggle. The dataset contains 43 sound recordings with varying duration ranging from 7 seconds to 90 seconds. Each recording is formatted into mono-pcm and the sample rate is synchronized to 44100hz. The dataset is split into two versions with unique corresponding feature, namely Chromagram and Mel Frequency Cepstrum Coefficient (MFCC). To ensure compatibility and prevent shape errors, the feature frames are clipped to match the length of the shortest frame in the dataset.

3.2. Graph Representation

We created two graphs: one containing MFCC spectrogram and the other containing Chromagram spectrogram. Both graphs have 43 nodes, each node containing label, filename, and either MFCC or Chromagram data. Utilising DTW, each node's feature is compared to all other nodes in graph. Once the shortest distance for that node is found, a weighted edge is constructed between the corresponding nodes. The edge weight is determined from distance obtained without any additional process. The node which has found its best pair is then removed from the comparison pool. Fig 2 shows a graph made from each feature.

Let $G = (N, E)$ be a graph where N is the set of nodes and E is the set of edges. The formula for creation of edges can be denoted as $E = (a, b) | a, b \epsilon N, a \neq b, f(a, b) = \min_{b' \epsilon N \setminus \{a\}} f'(a, b')$ where a and b are nodes in N. The function f' takes two nodes' attributes as Dynamic Time Warping inputs. The distance calculation is done for each node a with every other node b' within the set of nodes N that is not node a. The function f returns the distance between node a, and the other node with the shortest distance, b.



(a) Graph with shortest MFCC distance as edges

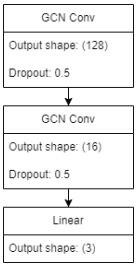
(b) Graph with shortest Chromagram distance as edges

Fig. 2: Graphs using MFCC and Chromagram

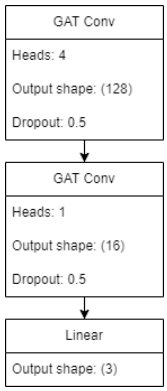
3.3. Models Training

In this stage, we have two versions of inputs ready. The first one is two graphs that contain node attributes as well as weighted edges. The second one is MFCC list and chromagram list before graph representation's process. The first one is used for training Graph Neural Networks. Namely GCN (Graph Convolutional Network)[9], GAT(Graph

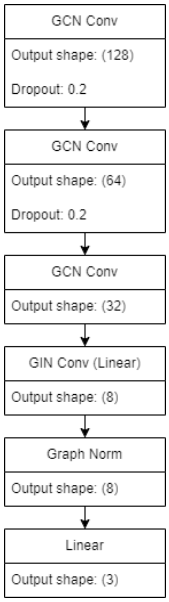
Attentional Network)[10], and GraphMix[11]. Meanwhile, the second input is used to fit LSTM(Long Short Term Memory), a type of RNN(Recurrent Neural Network). The models architecture are shown below in fig 3. The variable N in Fig.3.d denotes the different shapes of mfcc and chromagram version.



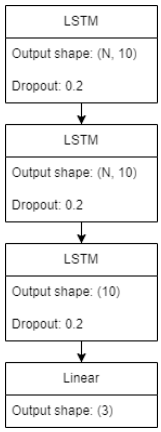
(a) Graph Convolutional Network



(b) Graph Attentional Network



(c) GraphMix



(d) Reccurent Neural Network (LSTM)

Fig. 3: Models Architecture

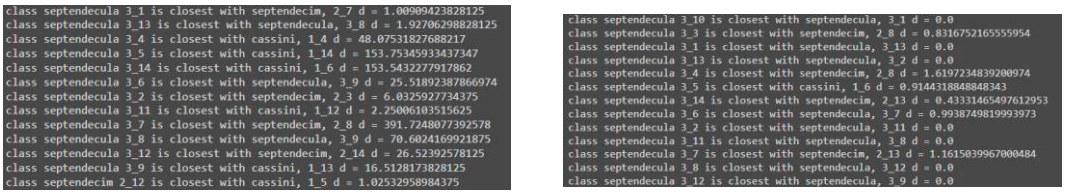
4. Result and Discussion

4.1. Result

Our main objective is to test the DTW-graph based approach effectiveness vs graph less with RNN. Secondly, we also prove how using chromagram instead of mfcc can eliminate denoising process. To achieve this, the experiment is done with no denoising and is compared to previous experiment which performed denoising on the same dataset.

Both features are experimented in both approaches such that we have chromagram vs mfcc effectiveness as feature in DTW distance computation. To measure DTW performance, not only will we look at metrics such as accuracy, loss but also perform analysis on the gap of both features distance output.

We used 75:25 train test ratio on 43 audio recordings and a learning rate of 0.001 for all models. The number of epochs that was used for GNN models are 300, while RNN is 50. GNN models are made using pytorch geometric library while RNN is made with keras. The experiment is done using Google Colab with regular CPU on Python 3. We instantiated each model twice to fit both feature input shapes.



(a) Distance on MFCCs (b) Distance on Chromagrams

Fig. 4: some samples on DTW distance on both features

Fig 4 shows some samples on how the distance obtained from pair of mfcc spectrograms differs from chromagram. Distance obtained from DTW on mfcc averaged 54.16 while chromagram averaged 0.75 . The min distance from 43 instances using mfcc is 1.03 and the max distance is 576.64. The min and max distance using chromagram are 0.0 and 6.65.

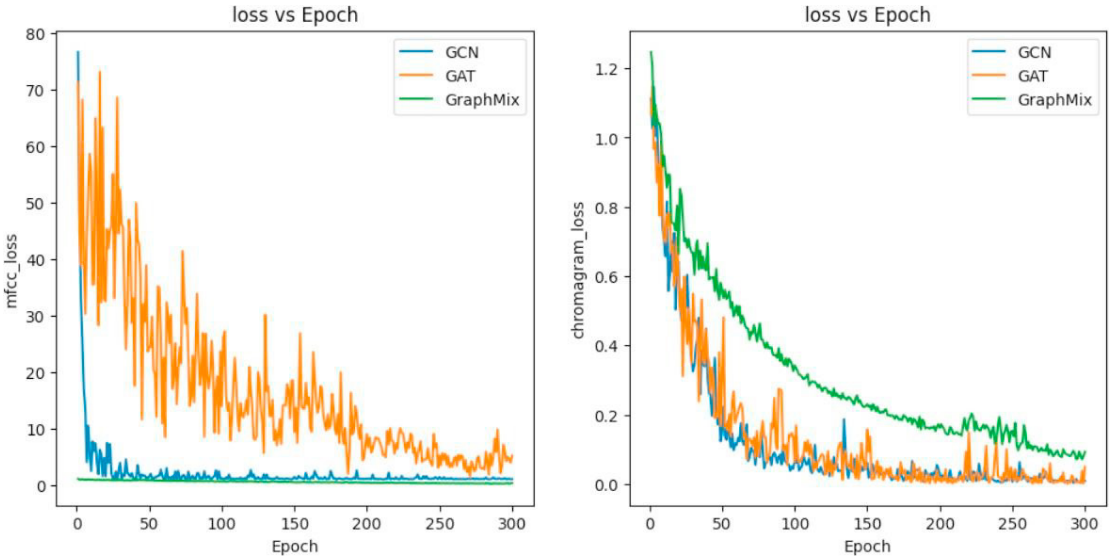


Fig. 5: Train loss of graph neural networks

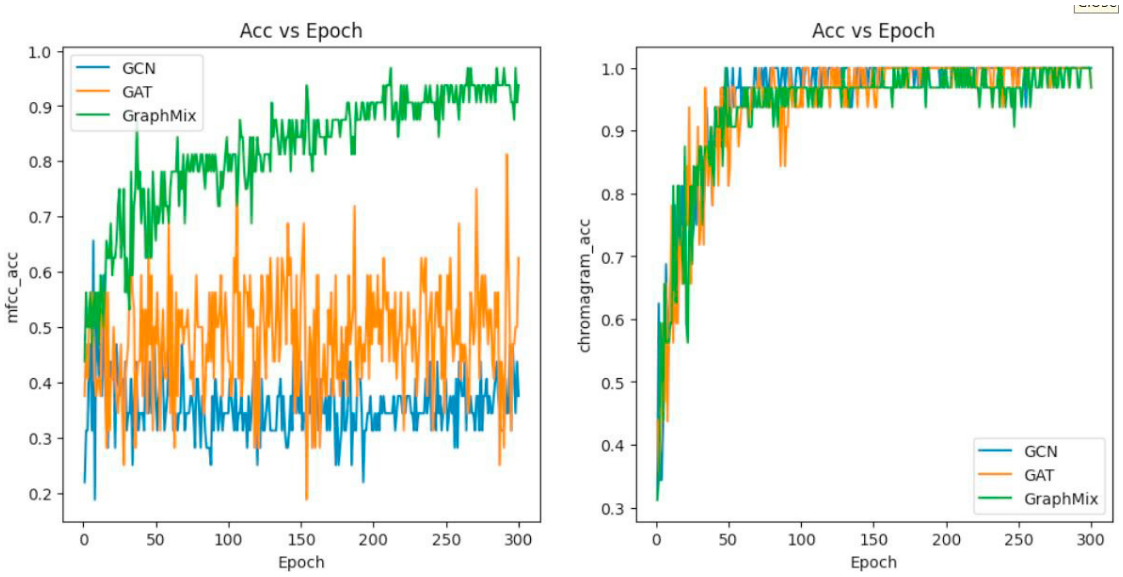


Fig. 6: Train Accuracy of GNN models

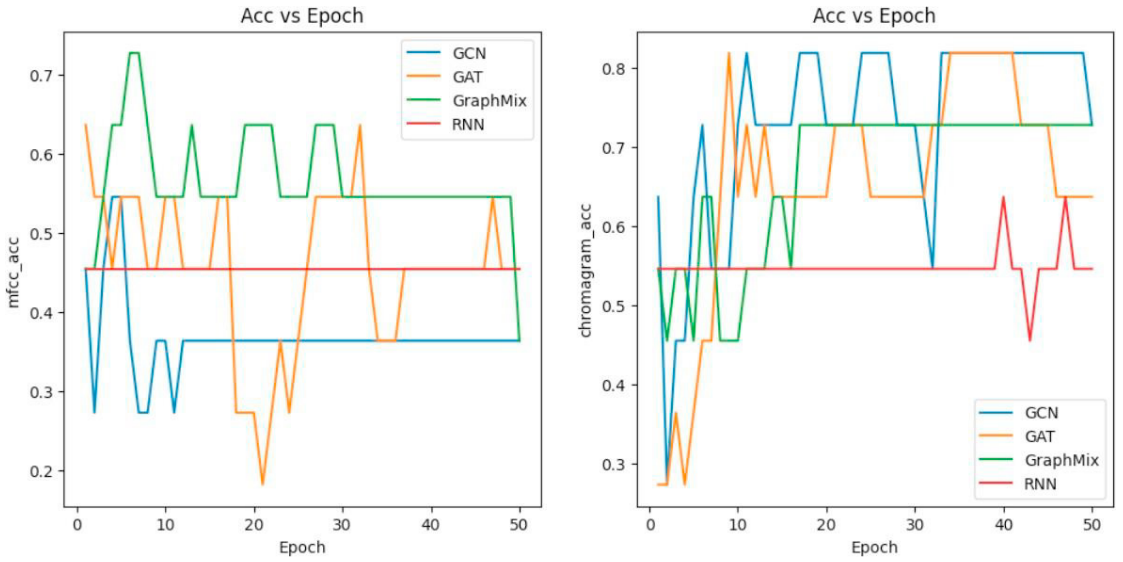


Fig. 7: Test Accuracy of all models

Table 1: Performances of all models

Graph	Feature	Model	Test Accuracy (%)
Yes	MFCC	GCN	54.55
		GAT	63.64
		GraphMix	72.73
No		RNN	45.45
Yes	Chromagram	GCN	81.82
		GAT	81.82
		GraphMix	81.82
No		RNN	62.50

The result could be seen in table 1. The graph approach using mfcc on GCN, GAT, GraphMix produced 54.55%, 63.64%, 72.73% accuracy. The loss of GCN, GAT, GraphMix with mfcc started on 76.69, 71.29, 1.07 as shown in fig 5. Fig 6 shows the GraphMix reaching a peak of 96.88% accuracy with an upward trend while GAT and GCN move sideway with peaks of 81.25% and 65.82% respectively. Fig 7 shows more detailed movement of test accuracy with mfcc. The x axis is trimmed to fit RNN's epoch parameter. RNN stagnated with 45.45% accuracy while the GNN models have best accuracies that can be seen in table 1.

The graph approach using chromagram on GCN, GAT, GraphMix reached the same test accuracy of 81.82%. The loss of GCN, GAT, GraphMix with chromagram started on 1.12, 1.06, 1.24 as shown in fig 5. Fig 6 shows all GNN models moving in an upward trend, all peaking on 100% train accuracy. The right side of fig 7 shows more detailed movement of test accuracy with chromagram. The best accuracies of RNN and GNN models are 62.50% and 81.82% respectively.

4.2. Discussion

As shown in Fig 6, GraphMix performed well irrespective of features. The addition of GIN Conv which have been proven to be the most powerful GNN by [12] and GraphNorm [13] is the reason of such result. Fig 5 shows how GraphNorm layer handled the loss better than other models. GNN models with DTW graph based approach achieved better accuracy than graph less approach using RNN.

From feature standpoint, chromagram feature delivered better result compared to mfcc. This is because we do not perform any denoising while other paper with MFCC and CNN did. Tey et al [2] performed denoising with binary thresholding and reached higher result with Ha¨rma¨ algorithm and CNN. Zhang et al .[4] with a different dataset of 93 recordings, preprocessed the data with pre-emphasis to balance the Signal-to-Noise ratio, . Both fig 4 and fig 5 shows the indication of mfcc poor perfomances. Fig 4 shows DTW on pair of mfcc return abnormally large number which is the total opposite of what chromagram produced. Fig 5 shows that GCN and GAT using mfcc started with a loss 70 times higher than with chromagram. The similar accuracies on chromagram graph models indicates that a large effort on model architecture creation is not needed. This implies that the graph constructed from DTW with chromagram is efficient enough. This concludes that we can eliminate the denoising process altogether by choosing chromagram over mfcc as feature, proving [7] research result on chromagram's robustness to noises. With this, we can potentially lower computational process in pre-processing area.

However, we have some disadvantages with previous researches, Dokania et al. [5] achieved 99.5% accuracy with node level classification graph-based approach on 1000 music recordings. One issue that arises when working with small dataset, is how one wrong prediction in test set cost us several percent of accuracy. Greater number of data like Dokania's prevent the issue from happening. Furthermore, bigger number introduce better representation of what the data generally is. On the same dataset, Tey et al [2] provided results on full length dataset and cut dataset. The full length dataset is the one that our research mainly focused on. The cut dataset used Ha¨rma¨ al- gorithm which extract individual syllables resulting in 684 samples. Moreover, the lengthy step that previous research done in removing noises explained why graph less approach with RNN performed badl .The problem lies within the noise in the dataset. Therefore a way to represent the data clearly is needed. The limitation of this research scope may not be even in comparations. Nevertheless, our experiment shows that the lengthy steps taken by previous research to remove the noises can be reduced by selecting chromagram as the feature. Moreover, the DTW graph based approach vs graph

less with chromagram shows a gap of ±19% accuracy. This concludes that graph based approach is one of the better data representation and should be taken into consideration in future research.

5. Conclusion

We conducted a research on cicada audio dataset containing 43 recordings. We evaluated a graph-based approach compared to other graph-less approach commonly used in time series. We also compared effectiveness of two unique features in noisy dataset. DTW is used to provide information on neighboring mfcc/chromagram relationships. Overall chromagram performed better than mfcc in all cases. Using chromagram, we made a graph representation powerful enough to outperform graph-less approach. We have seen similar method [5] used on 1000 sample attaining 99.5% accuracy. We hope this research can further boost the advancement of graph neural networks and its surrounding methods.

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